

Multiple Choice (10 marks)

1. Let `l = [1,2,3,4]`. What is `sum(l)` in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
2. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) `**`
 - (b) `//`
 - (c) `is`
 - (d) `not in`
3. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
4. Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][1]` evaluates to 0.
7. True or False: In Python 3, the function `max(list)` returns the item with the highest value from the given list.
8. True or False: The function $O(\log \log N)$ grows slower than $O(\log N)$ and $O(N)$.
9. True or False: The value of the Python expression `4 + 3 % 2` is 5.
10. True or False: `"a" + "ab"` in Python 3 produces an error due to incompatible data types.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to generate all sublists of a given list.
12. Write a function to remove a specified column from a given nested list.

End of Paper